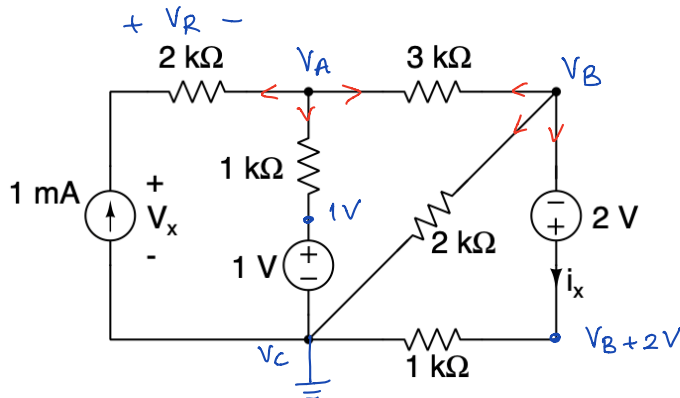


EEE 211 - SPRING 2022
MIDTERM EXAM SOLUTIONS

Question 1 [20 Points]:

- (a) Analyze the circuit given below using nodal analysis. Clearly label the node voltages (as V_A , V_B , V_C , ...) and mark the ground node. Find all the node voltages.
- (b) Using the result in part (a), find the current i_x and the voltage V_x as marked in the circuit.



(12) a) KCL at V_A : $-1 \text{ mA} + \frac{V_A - 1}{1 \text{ k}} + \frac{V_A - V_B}{3 \text{ k}} = 0$

$$4V_A - V_B = 6$$

KCL at V_B : $\frac{V_B - V_A}{3 \text{ k}} + \frac{V_B}{2 \text{ k}} + \frac{V_B + 2}{1 \text{ k}} = 0$

$$11V_B - 2V_A = -12 \quad \times 2$$

$$22V_B - 4V_A = -24$$

$$+ 4V_A - V_B = 6$$

$$21V_B = -18 \Rightarrow$$

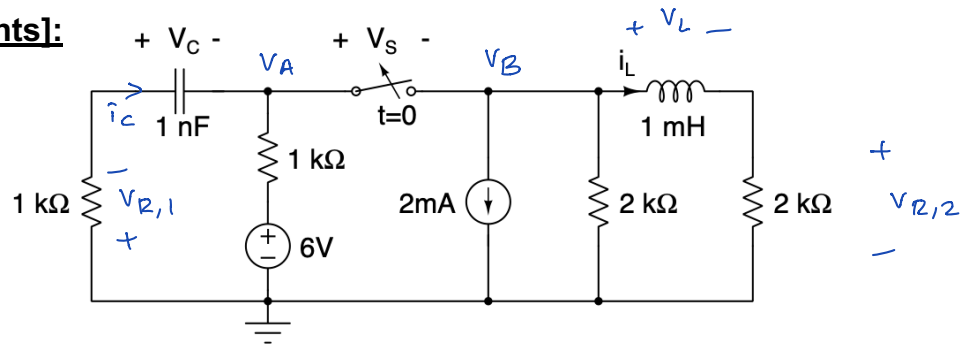
$$V_B = -\frac{6}{7} \text{ V}$$

$$V_A = \frac{V_B + 6}{4} = \frac{-\frac{6}{7} + 6}{4} = \frac{9}{7} \text{ V}$$

(8) b) $i_x = \frac{V_B + 2}{1 \text{ k}} = \frac{-\frac{6}{7} + 2}{1 \text{ k}} = \frac{8}{7} \text{ mA}$

From KVL: $V_x = V_R + V_A = 1 \text{ mA} \cdot 2 \text{ k} \Omega + \frac{9}{7} = 2 + \frac{9}{7} = \frac{23}{7} \text{ V}$

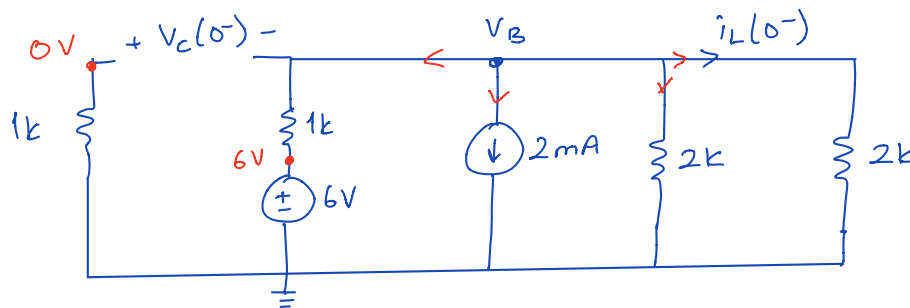
Question 2 [30 points]:



In the circuit above, the switch is kept closed (i.e., conducting) for $t < 0$ and is opened at $t = 0$.

- Find $v_c(0^-)$ and $i_L(0^-)$, i.e., right before the switch is opened.
- Find and plot $v_c(t)$ for $t \geq 0$. Mark important time points and voltages.
- Find and plot $i_L(t)$ for $t \geq 0$. Mark important time points and currents.
- Using the results in (b) and (c), find the voltage across the switch, $v_s(t)$, for $t \geq 0$.

(8) a) switch is conducting for a long time: $C \rightarrow$ open, $L \rightarrow$ short



KCL at V_B : $\frac{V_B - 6}{1k} + 2mA + \frac{V_B}{2k} + \frac{V_B}{2k} = 0 \Rightarrow V_B = 2V$

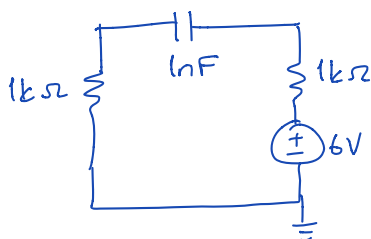
Then, $v_c(0^-) = 0 - V_B = \boxed{-2V}$

$i_L(0^-) = \frac{V_B}{2k} = \frac{2V}{2k} = \boxed{1mA}$

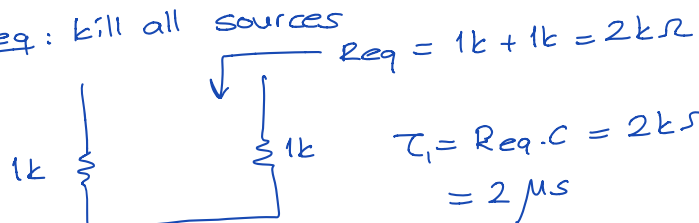
(8) b) After $t > 0$, switch is open (not conducting).
The two sides of the circuit are isolated from each other.

* For capacitor:

$V_{init} = v_c(0^-) = -2V$

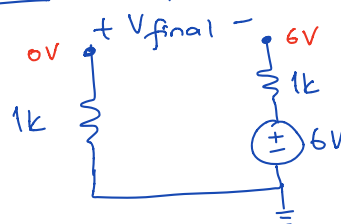


Req: kill all sources



$\tau_1 = R_{eq} \cdot C = 2k\Omega \cdot 1nF = 2\mu s$

V_{final} : Capacitor acts like open circuit

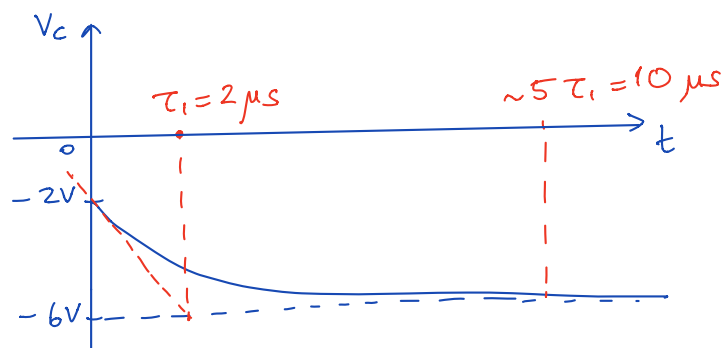


$V_{final} = 0 - 6V = -6V$

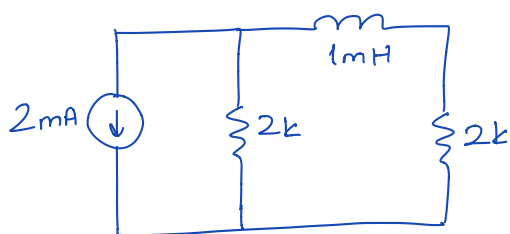
$$V_C(t) = V_{final} + (V_{init} - V_{final})e^{-t/\tau_1}$$

$$= \boxed{-6 + 4e^{-t/\tau_1} \text{ V}}$$

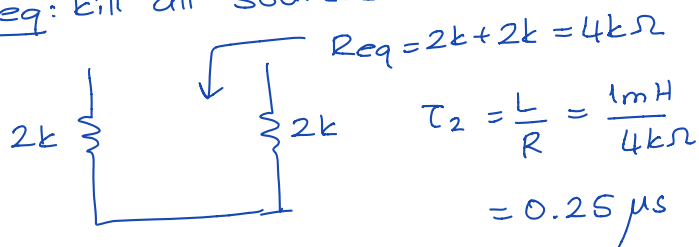
for $t \geq 0$, $\boxed{\tau_1 = 2\mu\text{s}}$



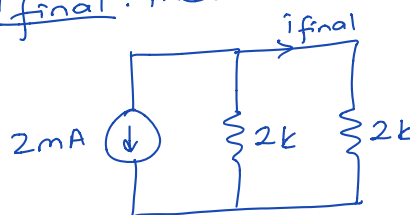
(8) c) For inductor:
 $i_{init} = i_L(0^-) = 1\text{mA}$



Req: kill all sources



i_{final} : inductor acts like short circuit

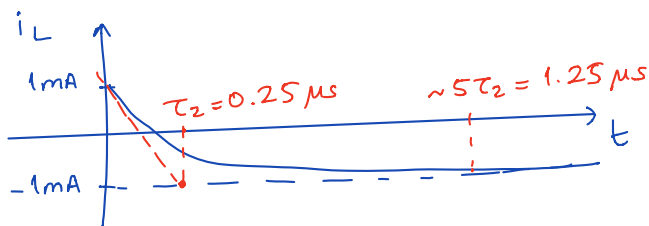


$$i_{final} = -2\text{mA} \cdot \frac{2k}{2k+2k} = -1\text{mA}$$

$$i_L(t) = i_{final} + (i_{init} - i_{final})e^{-t/\tau_2}$$

$$= \boxed{-1 + 2e^{-t/\tau_2} \text{ mA}},$$

for $t \geq 0$, $\boxed{\tau_2 = 0.25\mu\text{s}}$



(6) d) KCL at V_A : $\frac{V_A + V_C}{1k} + \frac{V_A - 6}{1k} = 0$

$$V_A = \frac{6 - V_C}{2} = \frac{6 - (-6 + 4e^{-t/\tau_1})}{2} = 6 - 2e^{-t/\tau_1} \text{ V}$$

KCL at V_B : $2\text{mA} + \frac{V_B}{2k} + i_L = 0$

$$V_B = -4 - i_L \cdot 2k = -4 - (-1 + 2e^{-t/\tau_2}) \cdot 2 = -2 - 4e^{-t/\tau_2} \text{ V}$$

Then,

$$V_S(t) = V_A - V_B = \boxed{8 - 2e^{-t/\tau_1} + 4e^{-t/\tau_2} \text{ V}}$$

Alternative Solution for (d):

$$V_S = V_A - V_B = (-V_C - V_{R,1}) - (V_L + V_{R,2}) = -V_C - i_C \cdot 1k - V_L - i_L \cdot 2k$$

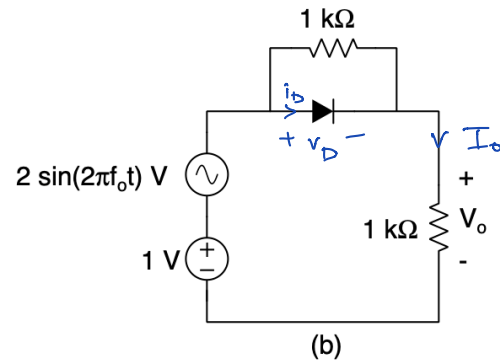
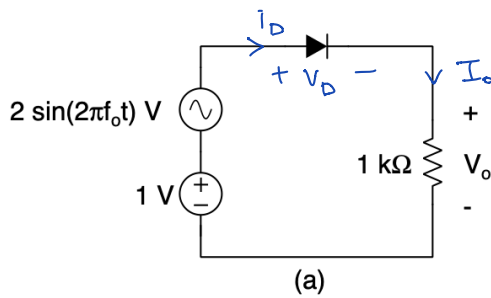
$$i_C = C \frac{dV_C}{dt} = 10^{-9} \cdot \left(-\frac{4}{\tau_1} e^{-t/\tau_1} \right) = \frac{-4 \cdot 10^{-9}}{2 \cdot 10^{-6}} \cdot e^{-t/\tau_1} = -2 e^{-t/\tau_1} \text{ mA}$$

$$V_L = L \frac{di_L}{dt} = 10^{-3} \left(-\frac{2}{\tau_2} e^{-t/\tau_2} \cdot 10^{-3} \right) = \frac{-2 \cdot 10^{-6}}{0.25 \times 10^{-6}} e^{-t/\tau_2} = -8 e^{-t/\tau_2} \text{ V}$$

$$V_S(t) = 6 - 4 e^{-t/\tau_1} + 2 e^{-t/\tau_1} + 8 e^{-t/\tau_2} + 2 - 4 e^{-t/\tau_2}$$

$$= \boxed{8 - 2 e^{-t/\tau_1} + 4 e^{-t/\tau_2} \text{ V}}$$

Question 3 [25 points]:



For the circuits shown above, assume that $V_{ON} = 0.6 \text{ V}$ for the diodes and $f_0 = 100 \text{ Hz}$.

- (a) Find $v_o(t)$ for the circuit in (a). Plot $v_o(t)$ for $0 \leq t \leq T$, where $T = 1/f_0$. Mark important voltages and diode turn on/off time points. Specify the state of the diode for each time interval.
- (b) Find $v_o(t)$ for the circuit in (b). Plot $v_o(t)$ for $0 \leq t \leq T$, where $T = 1/f_0$. Mark important voltages and diode turn on/off time points. Specify the state of the diode for each time interval.

$$T = \frac{1}{100} = 0.01 \text{ s} = 10 \text{ ms}$$

(11) a) * when diode OFF $\Rightarrow i_D = I_o = 0 \Rightarrow V_o = I_o \cdot 1k = 0$

* when diode ON $\Rightarrow V_D = 0.6 \text{ V}$

$$V_o = 2 \sin(2\pi f_0 t) + 1 - 0.6 = 2 \sin(2\pi f_0 t) + 0.4 \text{ V}$$

Condition for diode to be ON:

$$i_D = I_o = \frac{V_o}{1k} = \frac{2 \sin(2\pi f_0 t) + 0.4 \text{ V}}{1k} > 0$$

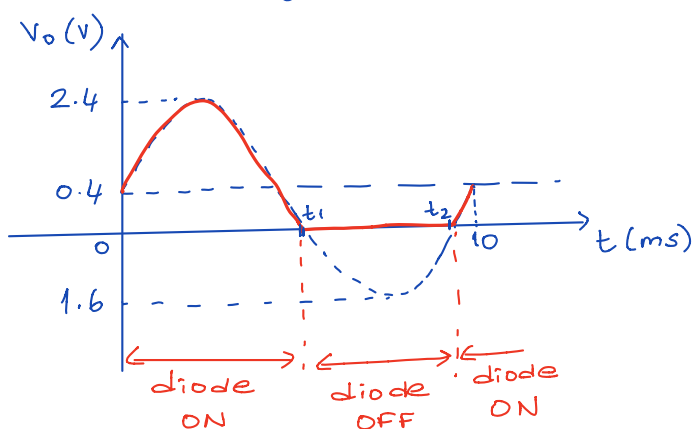
$$\sin(2\pi f_0 t) > -0.2 \Rightarrow t_0 = \frac{-\arcsin(0.2)}{2\pi f_0} = -0.32 \text{ ms}$$

$$t_0 + NT < t < \frac{T}{2} - t_0 + NT, \text{ for } N \in \mathbb{Z} \text{ (integer)}$$

* SO,

$$V_o(t) = \begin{cases} 2 \sin(2\pi f_0 t) + 0.4 \text{ V} & , -0.32 + N \cdot 10 < t < 5.32 + N \cdot 10 \\ 0 & , \text{ else} \end{cases}$$

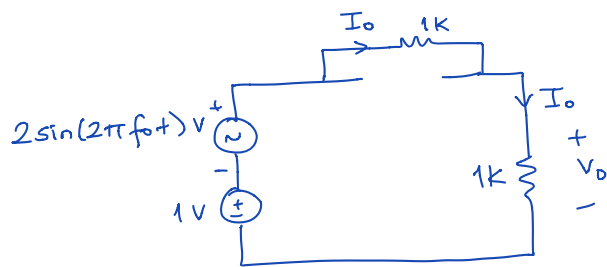
where t is in ms.



$$t_1 = \frac{T}{2} - t_0 = 5 + 0.32 = 5.32 \text{ ms}$$

$$t_2 = T + t_0 = 10 - 0.32 = 9.68 \text{ ms}$$

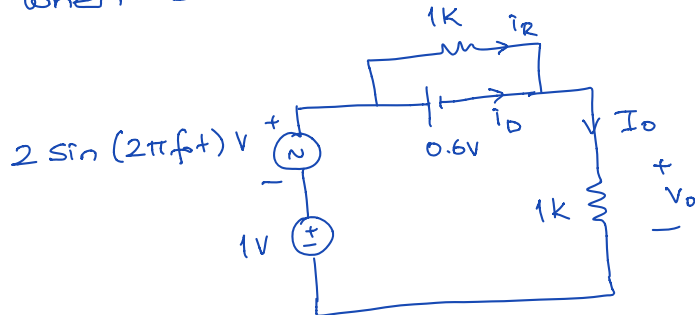
(14) b) * when diode OFF : $i_D = 0$



$$V_o = \frac{2\sin(2\pi ft) + 1}{2} \quad (\text{voltage divider})$$

$$= \sin(2\pi ft) + 0.5V$$

* when diode ON : $V_D = 0.6V$



$$\text{KVL: } V_o = 2\sin(2\pi ft) + 1 - 0.6$$

$$= 2\sin(2\pi ft) + 0.4V$$

Condition for diode to be ON:

$$i_D = I_o - i_R > 0 \Rightarrow \frac{V_o}{1k} - \frac{0.6}{1k} > 0$$

$$2\sin(2\pi ft) + 0.4 > 0.6$$

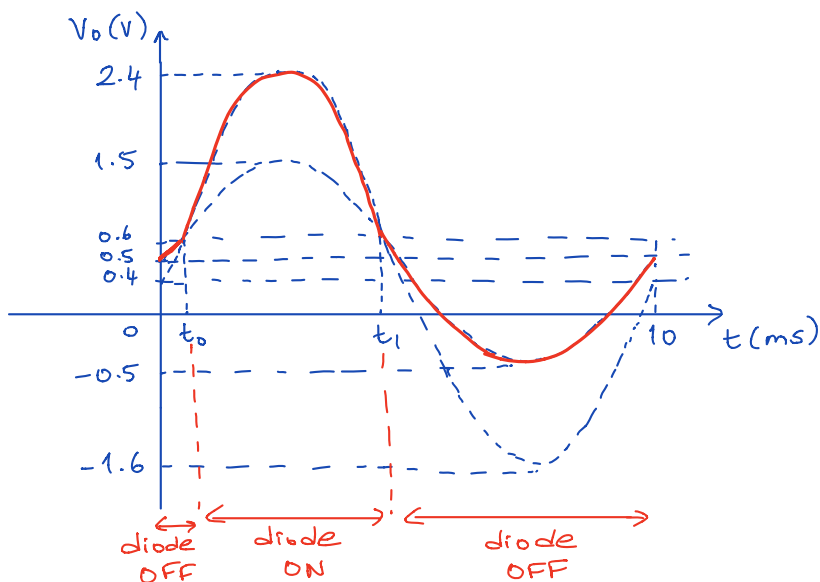
$$\sin(2\pi ft) > 0.1 \Rightarrow t_0 = \frac{\arcsin(0.1)}{2\pi f_0} = 0.16ms$$

$$t_0 + NT < t < \frac{T}{2} - t_0 + NT, \text{ for } N \in \mathbb{Z} \text{ (integer)}$$

* So,

$$V_o(t) = \begin{cases} 2\sin(2\pi ft) + 0.4V, & 0.16 + N \cdot 10 < t < 4.84 + N \cdot 10 \\ \sin(2\pi ft) + 0.5V, & \text{else} \end{cases}$$

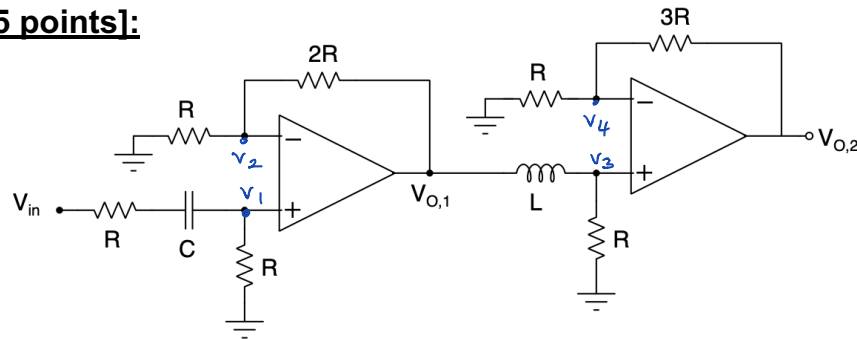
where t is in ms



$$t_0 = 0.16ms$$

$$t_1 = \frac{T}{2} - t_0 = 4.84ms$$

Question 4 [25 points]:



For the circuit given above, assume that $v_{in}(t)$ is sinusoidal, the OPAMPs are ideal and not saturated, $R = 1 \text{ k}\Omega$, $C = 5 \text{ nF}$, $L = 1 \text{ mH}$.

- (a) Find the transfer function $H_1(\omega) = \frac{V_{o,1}}{V_{in}}$ in parametric form (i.e., do not insert component values). Express the transfer function in one of the following standard forms:

$$A \frac{1}{1+jX} \quad , \quad A \frac{jX}{1+jX} \quad , \quad A \frac{1+jX}{1+jY}$$

- (b) What type of filter is $H_1(\omega)$? Find the 3-dB cut-off frequency in units of Hz. Find the maximum gain in dB.
- (c) Find the transfer function $H_2(\omega) = \frac{V_{o,2}}{V_{o,1}}$ in parametric form (i.e., do not insert component values). Express the transfer function in one of the standard forms given in part (a).
- (d) What type of filter is $H_2(\omega)$? Find the 3-dB cut-off frequency in units of Hz. Find the maximum gain in dB.
- (e) Find the transfer function $H(\omega) = \frac{V_{o,2}}{V_{in}}$.
- (f) Find $v_{o,1}(t)$ and $v_{o,2}(t)$, if $v_{in}(t) = 3 \cos(2 \times 10^5 t + \pi/3) \text{ V}$.

(5) a) $V_1 = V_{in} \cdot \frac{R}{2R + \frac{1}{j\omega C}} = V_{in} \cdot \frac{j\omega RC}{1 + j2\omega RC}$ (voltage divider)

$V_2 = V_{o,1} \cdot \frac{R}{R + 2R} = \frac{V_{o,1}}{3}$ (voltage divider)

$V_1 = V_2 \Rightarrow V_{in} \cdot \frac{j\omega RC}{1 + j2\omega RC} = \frac{V_{o,1}}{3}$

$H_1(\omega) = \frac{V_{o,1}}{V_{in}} = \frac{3j\omega RC}{1 + j2\omega RC} = \left[\frac{3}{2} \cdot \frac{j2\omega RC}{1 + j2\omega RC} \right]$ in standard form

(4) b) $|H_1(\omega)| = \frac{3}{2} \frac{2\omega RC}{\sqrt{1 + 4\omega^2 R^2 C^2}} \Rightarrow \begin{cases} |H_1(0)| = 0 \\ |H_1(\omega \rightarrow \infty)| = \frac{3}{2} \end{cases} \Rightarrow \boxed{\text{HPF}}$

$\max |H_1(\omega)| = \frac{3}{2} \Rightarrow 20 \log_{10} \frac{3}{2} = \boxed{3.52 \text{ dB}}$

$|H(\omega_c)| = \frac{\max |H_1(\omega)|}{\sqrt{2}} = \frac{3}{2} \cdot \frac{1}{\sqrt{2}} = \frac{3}{2} \frac{2\omega_c RC}{\sqrt{1 + 4\omega_c^2 R^2 C^2}} \Rightarrow 2\omega_c RC = 1$

$\omega_c = \frac{1}{2RC} \Rightarrow f_c = \frac{1}{2\pi \cdot 2RC} = \frac{1}{4\pi \cdot 10^3 \cdot 5 \cdot 10^{-9}} = \boxed{15.92 \text{ kHz}}$

(5) c) $V_3 = V_{0,1} \cdot \frac{R}{R+j\omega L}$ (voltage divider)

$V_4 = V_{0,2} \cdot \frac{R}{R+3R} = \frac{V_{0,2}}{4}$ (voltage divider)

$V_3 = V_4 \Rightarrow V_{0,1} \cdot \frac{R}{R+j\omega L} = \frac{V_{0,2}}{4}$

$H_2(\omega) = \frac{V_{0,2}}{V_{0,1}} = \frac{4R}{R+j\omega L} = \boxed{4 \cdot \frac{1}{1+j\omega \frac{L}{R}}}$ in standard form

(4) d) $|H_2(\omega)| = 4 \cdot \frac{1}{\sqrt{1+\omega^2 \frac{L^2}{R^2}}} \Rightarrow \begin{matrix} |H_2(0)| = 4 \\ |H_2(\omega \rightarrow \infty)| = 0 \end{matrix} \left. \vphantom{\begin{matrix} |H_2(0)| = 4 \\ |H_2(\omega \rightarrow \infty)| = 0 \end{matrix}} \right\} \boxed{\text{LPF}}$

$\max |H_2(\omega)| = 4 \Rightarrow 20 \cdot \log_{10} 4 = \boxed{12 \text{ dB}}$

$|H_2(\omega_c)| = \frac{\max |H_2(\omega)|}{\sqrt{2}} = 4 \cdot \frac{1}{\sqrt{2}} = 4 \cdot \frac{1}{\sqrt{1+\omega_c^2 \frac{L^2}{R^2}}} \Rightarrow \omega_c \frac{L}{R} = 1$

$\omega_c = \frac{R}{L} \Rightarrow f_c = \frac{\omega_c}{2\pi} = \frac{R}{2\pi \cdot L} = \frac{10^3}{2\pi \cdot 10^{-3}} = \boxed{159.15 \text{ kHz}}$

(2) e) $H(\omega) = \frac{V_{0,2}}{V_{in}} = \frac{V_{0,1}}{V_{in}} \cdot \frac{V_{0,2}}{V_{0,1}} = H_1(\omega) \cdot H_2(\omega)$

$= \frac{3}{2} \frac{j2\omega RC}{1+j2\omega RC} \cdot 4 \cdot \frac{1}{1+j\omega \frac{L}{R}} = \boxed{6 \cdot \frac{j2\omega RC}{(1+j2\omega RC)(1+j\omega \frac{L}{R})}}$

(5) f) $\omega_0 = 2 \cdot 10^5 \text{ rps}$, $V_{in} = 3e^{j\pi/3} \text{ V}$

$H_1(\omega_0) = \frac{3}{2} \cdot \frac{j2 \cdot 2 \cdot 10^5 \cdot 10^3 \cdot 5 \cdot 10^{-9}}{1+j2 \cdot 2 \cdot 10^5 \cdot 10^3 \cdot 5 \cdot 10^{-9}} = \frac{3}{2} \frac{j2}{1+j2} = \frac{3e^{j\pi/2}}{\sqrt{5}e^{j\arctan(2)}}$

$V_{0,1} = V_{in} \cdot H_1(\omega_0) = 3e^{j\pi/3} \cdot \frac{3}{\sqrt{5}} e^{j(\pi/2 - \arctan(2))} = 4.02 e^{j1.51}$

$V_{0,1}(t) = 4.02 \cos(2 \cdot 10^5 t + 1.51) \text{ V}$

$H_2(\omega) = \frac{4}{1+j2 \cdot 10^5 \cdot \frac{10^{-3}}{10^3}} = \frac{4}{1+j0.2} = \frac{4}{\sqrt{1.04}} e^{j\arctan(0.2)}$

$V_{0,2} = V_{0,1} \cdot H_2(\omega) = 4.02 e^{j1.51} \cdot \frac{4 \cdot e^{-j\arctan(0.2)}}{\sqrt{1.04}} = 15.7 e^{j1.31}$

$V_{0,2}(t) = 15.7 \cos(2 \cdot 10^5 t + 1.31) \text{ V}$